Jaewon Kim

Undergraduate Student, Korea University

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RESEARCH INTERESTS

My research lies at the intersection of **deep reinforcement learning** and **generative models**, with the goal of building expressive and scalable algorithms for the real world. I am particularly interested in **physics-informed machine learning**, which uses PDEs and stochastic processes to guide learning, and its applications to **robotics** and **AI for science**.

EDUCATION

Korea University **Mar 2022 – Expected Aug 2028**

B.S. in Semiconductor Engineering *Seoul, Korea*

- **GPA**: 4.35/4.5 (3.93/4.0)
- Full tuition and stipend from the SK Hynix-KU Partnership Scholarship.
- Leave of absence for mandatory military service (Mar 2025 – Dec 2026).

University of Texas at Austin **Jan 2027 – Jun 2027**

Exchange Student, Department of Mathematics *Austin, TX*

- Supported by the Korea-U.S. Advanced Technology Youth Exchange Support Program (KIAT).

RESEARCH EXPERIENCE

Complex Data Analytics Lab (CDAL), Korea University **Feb 2024 – Present**

Undergraduate Researcher (Advisor: Prof. Junhee Seok) *Seoul, Korea*

High-Dimensional Continuous Conditional Generation [P1]

- Led the theoretical formulation and algorithm design of `HYPE-CCDM`, extending continuous conditional diffusion from scalar to multi-dimensional regression labels.
- Addressed distance concentration and label sparsity in high dimensions by reformulating the vicinal objective with random 1D projections and distance-aware prioritized sampling, achieving state-of-the-art results.

Physics-Informed Policy Learning [J1]

- Proposed Fokker-Planck Soft Actor-Critic (`FP-SAC`), recasting soft policy improvement as steering the policy toward the stationary distribution of a Langevin SDE.
- Mitigated mode collapse of reverse-KL updates by training normalizing-flow policies through Fokker-Planck PDE residual minimization, yielding expressive multi-modal policies.

PUBLICATIONS

(P: Preprint, J: Journal, *: Equal contribution, †: Equal advising)

[**P2**] **NanoDiff: Physics-Guided Diffusion for Image-Based Nanophotonic Device Design**
Insoo Kim, Jaewon Kim, Jangwon Seo, Junhee Seok†, Bohyung Han†
Under Review

[**P1**] **Scalable Continuous Conditional Diffusion via Random Projections and Prioritized Sampling**
Jaewon Kim*, Hyo-Seok Hwang*, Junhee Seok
Under Review

[**J1**] **Fokker-Planck Soft Actor-Critic**
Hyo-Seok Hwang, Jaewon Kim, Junhee Seok
IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2026

HONORS & SCHOLARSHIPS

Korea-U.S. Advanced Technology Youth Exchange Scholarship (∼$9,000) **Nov 2026 – Present**
Korea Institute for Advancement of Technology (KIAT)

SK Hynix-KU Partnership Scholarship (Full tuition and stipend) **Spring 2022 – Present**
SK Hynix and Korea University

KU Alumni Scholarship (∼$8,000) **Spring 2022 – Fall 2024**
Korea University Alumni Association

Outstanding Poster Award **2024**
12th SK Hynix Academic Conference

Semester High Honors **Spring 2022 – Fall 2024**
Korea University

Dean's List (GPA 4.5/4.5) **Spring 2022**
Korea University

ACTIVITIES

Young Engineers' Honor Society (YEHS), NAEK **Mar 2024 – Present**
Selected as a member of the honor society of the National Academy of Engineering of Korea.

BITAmin, Intercollegiate ML/DL Study Group **Aug 2023 – Aug 2024**
Studied ML/DL papers weekly and carried out team projects.

Korea University International Student Assistant (KUISA) **Feb 2023 – Feb 2024**
Helped incoming international students settle in at Korea University.

SKILLS

Programming: Python (PyTorch, TensorFlow), MATLAB, C, C++
Languages: Korean (native), English (fluent, TOEFL iBT 118)

SELECTED COURSEWORK

Mathematics & Statistics: Engineering Mathematics I and II, Introduction to Linear Algebra, Probability and Random Process, Fundamentals of Data Science, Data Learning and Intelligence

Electrical & Computer Engineering: Signals and Systems, Digital Signal Processing, Digital Image Processing, Deep Signal Processing, Object-Oriented Programming, Computer Systems